



Embedded feature fusion for multi-view multi-label feature selection

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ARTICLE INFO

Keywords:

Multi-view learning
Multi-label learning
Feature selection
Feature fusion

ABSTRACT

With the explosive growth of data sources, multi-view multi-label learning (MVML) has garnered significant attention. However, the task of selecting informative features in MVML becomes more challenging as the dimensionality increase. Existing methods often extract information separately from the consensus part and the complementary part, potentially leading to noise attributed to ambiguous segmentation. In this paper, we propose an embedded feature selection model that combines with two aspects, which are the feature fusion between views and feature enhancement. Firstly, we calculate the adaptive weight of each view based on the local structure relations, and integrate it into one unified feature matrix. Subsequently, the mapping between unified feature matrix and ground-truth label matrix is established. Furthermore, a regularizer for the feature weight of each view is constructed to emphasize its characteristic, respectively. As a result, the relationship for inter-view and intra-view has been simultaneously considered, preserving comprehensive information of features by minimizing the difference between two types of feature weight. Experimental results demonstrate the superior performance of our method in coping with feature selection.

1. Introduction

Feature selection is a crucial pre-processing step in multi-label learning, particularly in scenarios involving rich label concepts. It has wide application in fields such as image annotation [1] and action recognition [2], where high-dimensional data and complex relationships are common. However, applying multi-label methods directly to multi-views with increasing dimensionality is not a prudent way. Data of high dimensionality can significantly increase the memory storage requirements, computational costs, and the number of noisy and redundant features in data analytics [3]. Moreover, the substantial dimensionality engenders challenges in accurately delineating inter-data relationships, thereby compromising information integrity and impeding the identification of pivotal features. As a result, the effective performance of feature selection in multi-view multi-label (MVML) learning remains a challenging task.

In the multi-view scenario, the pursuit of an equilibrium between consistency and complementarity through an exploration of inter-data relationships poses a thought-provoking question. Two fundamental observations exist in the multi-view data [4]. Firstly, the data from multiple sources often converge towards a consensus, indicating a consistent description of the labels. Secondly, due to the heterogeneous nature of the sources, each view contributes a distinctive portion, as depicted in Fig. 1. For instance, certain labels such as ‘Rain’ and

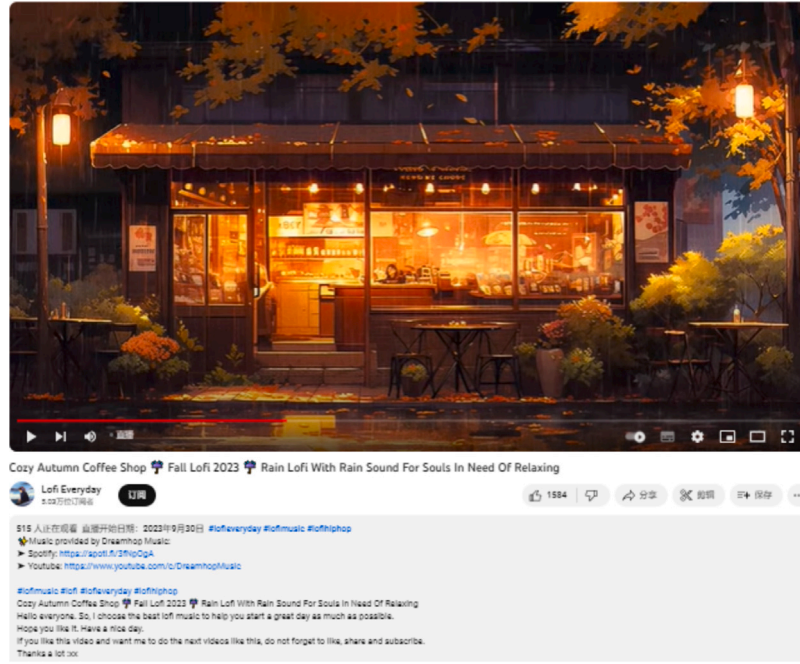
‘Relaxing’ may appear in all views as common labels, while ‘Coffee shop’ may be exclusive to some views but not others.

Regarding the consensus problem, it is noteworthy that the presence of rich semantics is particularly pronounced in comparison to a single view. Consequently, the integration of diverse data sources becomes a novel area of focus [5]. Zhang et al. [6] propose the method which involves decomposing the feature matrix of each view into a consensus multi-view representation. Meanwhile, it leverages the Hilbert–Schmidt independence criterion to enhance the dependence between different views. Similarly, Liu et al. [7] employ matrix decomposition of features to construct a label-dependent shared subspace. In contrast to [6], this shared subspace is further constrained by the local geometric structure of the label, rather than being limited to the basis matrix in each view. Consequently, it is inevitable that the distinct features may be overlooked, thereby influencing the performance of feature selection [8].

In light of the principle of complementarity, several investigations have achieved significant progress in unsupervised multi-view learning. Regarding the preservation of individual view characteristics, the term imposes constraints on local geometric among different views [9,10]. However, the utilization of label formation encounters inherent limitations within the unsupervised domain. While the design of the objective function may incorporate the label as a learning variable, the constraints are heightened by its ambiguous attributes [11].

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Labels: Rain, Relaxing, Coffee shop

Fig. 1. The framework of the proposed EF^2FS method. The specific labels are shown in underline, others are common labels.

Accordingly, to fully exploit the information of labels in MVML, Tan et al. [12] further acquire distinctive features for each view by establishing mappings with labels, subsequent to obtaining the shared subspace as indicated by [6,7]. If there is inconsistency in the assumed labels for each view, this mapping transforms into a bridge between each view and its specific label. It is imperative to define and constrain the segment among views to prevent overlaps or omissions. Hence, certain studies have not commenced from the mapping of a single view. Zhu et al. [13] iteratively learn feature weights based on the intrinsic relationship between features across all views and the label matrix. Based on [13], Zhang et al. [14] consider the noises in the label matrix and introduce a latent label space in the objective function. However, these methods still find it difficult to distinguish the unique information of each view and ensure that the information is not duplicated or ignored.

Consequently, we start to change the perspective from the focus on each individual view to address the aforementioned challenges from a global-view. Concurrently, we adopt the embedded methodology to formulate the objective function. Notably, three primary types of feature selection methods exist, encompassing filter methods, wrapper methods, and embedded methods [15]. Filter methods select features based on statistical analysis or relationships between features. The filter method such as FisherScore [16] selects features based on statistical analysis or relationships between features. While they typically exhibit computational efficiency, they do not incorporate feedback from the learning algorithm. Conversely, the wrapper method such as in [17] selects features by training the learning algorithm with diverse feature subsets, subsequently evaluating and selecting based on the performance of the algorithm. Nonetheless, this method is susceptible to overfitting and heightened computational demands. In contrast, the embedded method such as SFESA [18] seamlessly integrates feature selection into the training process, capitalizing on intrinsic characteristics of the algorithm for feature selection. Consequently, its efficacy lies in circumventing the repetitive execution of classifiers and examination of every feature subset. Owing to its inherent advantages, embedded methods have garnered considerable attention, with a majority of recent pertinent endeavors leveraging its benefits, thus substantiating our rationale for its selection.

To this end, a multi-view multi-label method named Embedded Feature Fusion for Multi-view Multi-label Feature Selection (EF^2FS) is proposed, which concatenate different features from each view at first. However, directly merging these multiple feature spaces into a unified feature space may overlook the inherent differences present in each view. To overcome this limitation, we introduce view-specific weights during the concatenation process. Then, the final feature matrix is derived subsequent to the interaction between global and local features. The main contributions of this work can be summarized as follows:

- The proposed method, EF^2FS , provides a novel perspective to perform feature selection by emphasizing the information integrity, including the fusion process and distinctive matrix solving process.
- The feature weight in EF^2FS is derived by integrating the outcomes from both the global and local perspectives. This method enables the consideration of inter-view relationships while preserving the unique attributes associated with each view.
- We introduce an alternating optimization algorithm with multiplicative update rules to solve the model effectively and efficiently. The experiments are conducted on six datasets to demonstrate the advantages of our method, and the results prove its efficiency and excellent over state-of-the-art related methods.

In the rest of this paper, Section 2 introduces the existing literature. Section 3 provides comprehensive explanations of our proposed method. Following this, Section 4 showcases the experimental results and provides a detailed analysis of the model. Finally, in Section 5, we draw conclusions based on our findings and discuss potential directions for future research.

2. Related work

2.1. Sparse multi-label feature selection

2.1.1. Loss regularizer

The general constitution for sparse multi-label feature selection can be categorized into two main components: loss regularizer and penalty

regularizer [19]. Loss regularizer primarily focuses on establishing the mapping between features and labels, taking advantage of the supervised nature of the methods [20,21]. The standard loss regularizer is represented as follows, and we take the Frobenious norm as the example:

$$\min_W \|XW - Y\|_F^2. \quad (1)$$

Other representative works, such as [22–24], incorporate additional constraints or explore latent feature matrices as a preliminary step. For instance, Liu et al. [23] propose the construction of a latent mapping to eliminate redundant information by leveraging the projection matrix. The loss regularizer can be written as:

$$\min_V \|(Y - P^T X)^T V\|_F^2. \quad (2)$$

Similarly, Gao et al. [24] decompose the feature matrix into two non-negative matrices and subsequently construct a shared correlation with the label matrix. The loss regularizer is denoted as:

$$\begin{aligned} \min_{V,Q} \|X - VQ^T\|_F^2 \\ s.t. \quad V, Q \geq 0 \end{aligned} \quad (3)$$

2.1.2. Penalty regularizer

When the loss regularizer in the objective function exhibits similarity, the constraints imposed by the penalty regularizer vary according to specific requirements. In the domain of multi-label problems, the presence of a label matrix emphasizes the criticality of effectively utilizing label information [25]. The fundamental objective is to obtain accurate feature weights by acquiring pure labels. Our method utilizing matrix factorization effectively mitigates noise in the data. Alternatively, other techniques involve learning a new variable derived from the original label matrix [26,27], or introducing a coefficient matrix that modifies the original matrix to create an enhanced label matrix [28,29]. Notably, these methods differ from matrix factorization in that they maintain the size of the original matrix. If computational complexity is not a primary concern, these methods are more suitable for scenarios with lower levels of noise. Additionally, Fan et al. [30] enforce constraints on the learning direction of feature weights by considering label correlation and cluster assignment. This strategy aims to achieve optimal clustering results for instances that exhibit similarity.

Specific to the feature weight directly for feature selection, there are three distinct methods that can be summarized: (1) different norms [31]. (2) matrix factorization [30,32]. (3) constraints after matrix operations [28,33]. The second and third is similar as the above, which often combine with the label information. The first case involves considering various norms for feature weight selection, as explored by Li et al. [31]. The $l_{2,1}$ -norm aims for group sparsity in the matrix, while the l_1 -norm emphasizes the sparsity of the entire matrix [34]. However, the l_1 -norm may encounter issues with zero weight values, while the $l_{2,1}$ -norm may overlook redundant information between features [35].

The emergence of new challenges arising from multiple views, notably dimension issues, complicates the direct application of existing algorithms to the multi-view scenario. However, certain studies [13,14] draw inspiration from design of feature weight in the multi-label scenario, adopting varied norms in their methods. Diverging from prior research, our method implements weight constraints from both the global and local view, balancing commonality and complementarity of the model.

2.2. Multi-view multi-label learning

Sparse learning techniques offer several advantages, including the ability to effectively handle high-order data relationships and mitigate the problem of overfitting. Additionally, these methods exhibit faster computation speeds. As a result, sparse learning methods are frequently employed in MVML learning.

In addressing the challenge of integrating information from multiple views, existing methods can be classified into three distinct types. The first type involves merging features, as demonstrated by Liu et al. [36]. Their method focuses on acquiring the shared subspace while enforcing a low-rank constraint. Similarly, Yin et al. [37] decompose the feature matrix within each view to extract the shared subspace and consider the contribution of each view to constrain the structure of the shared subspace. However, these methods often overlook the design of a regularizer to preserve distinctive information.

The second type, exemplified by Liu et al. [38], approaches integration from the perspective of the label matrix. This method establishes a mapping between the features and enhanced labels to facilitate the integration process. Similarly, in our method, we also consider label noise and the mapping of individual views. Additionally, we incorporate a global mapping to reduce feature redundancy and missing situations.

The final type of integration methods combines the aforementioned two types, incorporating both labels and features in the integration process. Zhu et al. [39] propose a two-step method, wherein they first construct a latent label matrix as the integration part and subsequently decompose the feature matrix from each view to obtain additional integration parts for the features. However, the existing methods for obtaining the comprehensive part are relatively limited, often relying on simple techniques such as direct concatenation [40].

3. The proposed method

3.1. Notations

The relevant symbols utilized are presented in Table 1. The loss function in this paper is to establish the mapping between the fusion features $X^f \in \mathbb{R}^{n \times d}$ and ground-truth labels $Y^g \in \{0, 1\}^{n \times c}$, where n , d and c denotes the number of instances, the number of features and the number of labels. Thus, the feature weight for all features $A \in \mathbb{R}^{d \times c}$ can be derived through the solution procedure of least square, facilitating the subsequent ordering of features for enhanced interpretability and analysis.

3.2. Problem formulation

By capitalizing on the benefits of sparse learning, the proposed method $E^{F^2}FS$ primarily aims to integrate the feature matrices from different views through a novel fusion method. In the context of feature weights, it is crucial to take into account the collective views and prevent the neglect of important features in each view. Consequently, the objective is to develop a method that considers inter-view relationships while preserving the distinctive characteristics of each view. Leveraging the previously defined notations, we formulate the objective function $F_{E^{F^2}FS}$ to collectively facilitate feature selection in the MVML scenario.

$$\begin{aligned} F_{E^{F^2}FS} = & F_{loss}(\cdot) + \alpha F_{p1}(\cdot) + \beta F_{p2}(\cdot) \\ & + \gamma F_{p3}(\cdot) + \delta F_{p4}(\cdot). \end{aligned} \quad (4)$$

In accordance with the structure of the objective function, the last four regularizers serve as penalty terms. The parameters α , β , γ and δ are employed to control the respective functions within the objective. $F_{loss}(\cdot)$ establishes the mapping between the features across all views and the corresponding labels. The first penalty regularizer, $F_{p1}(\cdot)$, functions as a noise filter by identifying and mitigating noisy labels. Subsequently, to expose the feature weight in each view, the least square is employed on the features from each view in conjunction with the ground-truth labels, denoted as $F_{p2}(\cdot)$. In order to capture the characteristics of local feature weights, the difference between global feature weights and local feature weights is represented by the function $F_{p3}(\cdot)$. Finally, the group sparsity of the global feature matrix is enforced through the constraint imposed by $F_{p4}(\cdot)$. Through the analysis above, the five regularizer comprising the $E^{F^2}FS$ objective function collectively contribute towards achieving the ultimate goal.

Table 1

Description of Notations.

Symbol	Definition
$x^{(i)}$	Feature matrix of i th view, $x^{(i)} \in \mathbb{R}^{n \times d(i)}$ where $d(i)$ denotes the number of features of i th view.
V	The number of views is V .
Y	Observed label matrix, $Y \in \{0, 1\}^{n \times c}$.
G	Latent label matrix, $G \in \mathbb{R}^{n \times k}$, where k is the number of clusters for labels.
B	The coefficient matrix with label cluster, $B \in \mathbb{R}^{c \times k}$.
$w^{(i)}$	Feature weight of i th view, $w^{(i)} \in \mathbb{R}^{d(i) \times c}$.
$S^{(i)}$	The affinity matrix of i th view, $S^{(i)} \in \mathbb{R}^{n \times n}$.
$L^{(i)}$	The graph Laplacian matrix of i th view, $L^{(i)} \in \mathbb{R}^{n \times n}$.
x_m^f	The m th instance in the fusion feature matrix X^f .
y_{mj}^g	The j th label in the ground-truth label matrix Y^g . If the m th instance for the j th label is annotated, then the element $y_{mj}^g = 1$.
$a^{(i)}$	The feature weight of i th view which is splitting from the feature matrix for all views A , and $a^{(i)} \in \mathbb{R}^{n \times d(i)}$.

3.2.1. Fusion feature construction

Given the feature matrix $x^{(i)}$ in each view, concatenating each view directly is not the appropriate way as it tends to overlook the distinctions between the views. Thus, in order to emphasize the relationship among views, it becomes necessary to calculate the contribution of each view to the global feature matrix. This can be achieved by drawing inspiration from the theory of similarity for local geometric structures, wherein the measurement of each view can be viewed as its contribution to all the views. The underlying premise of this theory is that the feature matrix in each view exhibits a similar local geometric structure to the label matrix. In other words, the fundamental assumption of this theory is that the pairwise similarity of instances in the feature matrix is approximately equivalent to the similarity observed in the label matrix.

Firstly, we compute the affinity matrix $S^{(v)}$ for the features in v th view, and the element in the matrix is defined as:

$$s_{ij}^{(v)} = \begin{cases} \exp(-\frac{\|x_{i.}^{(v)} - x_{j.}^{(v)}\|_2^2}{\sigma^2}) & \text{if } x_{i.}^{(v)} \in \mathcal{N}_q(x_{j.}^{(v)}) \text{ or } x_{j.}^{(v)} \in \mathcal{N}_q(x_{i.}^{(v)}), \\ 0 & \text{otherwise} \end{cases}, \quad (5)$$

herein, σ represents a predefined parameter, while $\mathcal{N}_q(x_{j.}^{(v)})$ signifies the set of q nearest neighbors of $x_{j.}^{(v)}$ and the similar meaning of $\mathcal{N}_q(x_{i.}^{(v)})$ can be inferred. Then, the formula based on the theory for the similarity of pairwise instances can be represented as:

$$\frac{1}{2} \sum_{i,j=1}^n s_{ij}^{(v)} \|y_{i.}^{(v)} - y_{j.}^{(v)}\|_2^2. \quad (6)$$

The formula implies that the similarity between pairwise instances in the feature matrix is inversely related to the difference between the two vectors of the label matrix. Hence, the formula incorporates a condition that enables the attainment of the minimum value. According to the theory of the graph Laplacian, the graph Laplacian matrix $L^{(v)} = A^{(v)} - S^{(v)}$, where $A^{(v)}$ is the diagonal matrix, whose elements are $a_{ii}^{(v)} = \sum_{j=1}^n s_{ij}^{(v)}$. Thus, Formula (6) can be transformed into Formula (7) as:

$$\text{Tr}(y^T (A^{(v)} - S^{(v)}) y) = \text{Tr}(y^T L^{(v)} y). \quad (7)$$

Regarding the non-negative property of the view weight and the constraint that the sum of the view weights for each view is equal to 1, the v th view weight c_i can be obtained as [37]:

$$c_i = \frac{1/\text{Tr}(y^T L^{(i)} y)}{\sum_{i=1}^V 1/\text{Tr}(y^T L^{(i)} y)}. \quad (8)$$

Combined with the view weight, the fusion matrix X^f can be obtained after the process of the concatenating:

$$X^f = [c_1 x^{(1)}, c_2 x^{(2)} \dots c_V x^{(V)}]. \quad (9)$$

Next, the finer-grained feature weights are adjusted based on the property of the least square between the fusion matrix and the label matrix. This adjustment serves as a loss regularizer for the objective function.

$$F_{\text{loss}}(\cdot) = \|X^f A - Y\|_F^2, \quad (10)$$

The calculation for the row of feature weight matrix A in Formula (10) indicates the importance of this feature to different labels, referring to all instances in fusion feature matrix. The feature weight matrix undergoes iterative updates as part of the optimization process for the method.

3.2.2. Feature enhancement

The penalty part of the objective function, which aims to enhance local features, consists of three distinct parts: label enhancement, distinctive feature enhancement, and sparsity enhancement for feature weight.

In the first part, the presumption concerning the ideal label matrix frequently proves difficult to actualize in the real-world. Consequently, we utilize matrix factorization techniques to discern latent semantic labels. The latent labels typically embody characteristics such as density, real-valued, and lower-dimensionality, thereby facilitating the establishment of a mapping between them and the features [41]. This process can be analogized to label clustering, where labels of c -dimensions are aggregated into k -dimensional label clusters. The relationship between latent labels and the original ones can be obtained through factorization. Instances that resist effective clustering are deemed as noise.

$$F_{p1}(\cdot) = \|Y - GB^T\|_F^2, \quad (11)$$

where B denotes the label correlations between the c labels and k new semantic labels. The latent label matrix G represents the ground-truth labels Y^g mentioned in this paper, and it is no more described in other places. Consequently, it is imperative that the matrix G adheres to the non-negativity characteristic. We transform the Formula (8) and (10) as:

$$c_i = \frac{1/\text{Tr}(G^T L^{(v)} G)}{\sum_{i=1}^V 1/\text{Tr}(G^T L^{(v)} G)}. \quad (12)$$

$$F_{\text{loss}}(\cdot) = \|X^f A - G\|_F^2. \quad (13)$$

In the case of the second part, it is crucial to emphasize the uniqueness of each view, which serves to compensate for any lack of consensus information. The process of enhancing distinctive features in this part is carried out in two phases. The first step involves capturing the characteristics of each view, akin to the terms described in [12] and [38]. This form of mapping proves advantageous in situations where certain views, despite their overall lesser significance, still exhibit prominent features associated with specific labels. The formulation of this process is as follows:

$$F_{p2}(\cdot) = \sum_{i=1}^V \|x^{(i)} w^{(i)} - G\|_F^2. \quad (14)$$

By integrating the local distinctive property for the feature weight of each view, the global feature weight obtained from the loss function draw the advantages of it that prevent the disregard of specific features. Consequently, the second step aims to minimize the discrepancy between feature weight for all views A and the feature weight for each individual view $x^{(i)}$. To achieve this, the matrix A is partitioned based

on the respective view scale, denoted as $a^{(i)}$, ensuring compatibility with the size of the feature weight specific to each view. The formula can be expressed as:

$$F_{p3}(\cdot) = \sum_{i=1}^V \|a^{(i)} - w^{(i)}\|_F^2. \quad (15)$$

For the third part, the sparsity is controlled by the $l_{2,1}$ -norm, which is defined as:

$$F_{p4}(\cdot) = \|A\|_{2,1} = \sum_{p=1}^d \sum_{q=1}^c a_{pq}^2. \quad (16)$$

The attribute of the $l_{2,1}$ -norm encompasses two distinct components. Firstly, the l_2 -norm is employed to measure the distance between features, allowing for an initial assessment of feature dissimilarity. Subsequently, the l_1 -norm is utilized to compute the summation of each row. Consequently, the effect for performing the $l_{2,1}$ -norm facilitates the attainment of row sparsity within the matrix [42,43]. When combined with the local distinctive features, the global feature matrix is enhanced, thereby culminating in the determination of feature ranking.

3.2.3. Representation for objective function

In the method EF²FS, the examination of the association between features and labels assumes a crucial role by incorporating various considerations. These considerations encompass the filtration of labels, identification of distinctive features, and the promotion of sparsity within rows. As a result, a more effective mapping is established. The overall method leverages the utilization of a shared representation, achieved through a fusion of views and regularization techniques. Furthermore, penalty functions are employed to enhance local features, ensuring a harmonious balance between common and complementary information. Consequently, the ultimate objective function for our methods can be formulated as:

$$\begin{aligned} \min_{A, G, B, w^{(i)}} & \|X^f A - G\|_F^2 + \alpha \|Y - GB^T\|_F^2 + \beta \sum_{i=1}^V \|x^{(i)} w^{(i)} - G\|_F^2 \\ & + \gamma \sum_{i=1}^V \|a^{(i)} - w^{(i)}\|_F^2 + \delta \|A\|_{2,1}. \end{aligned} \quad (17)$$

s.t. $G \geq 0, B \geq 0$

By imposing constraints on the variable, the label matrix is proceeded with non-negative matrix factorization. This necessitates that the matrix G remains non-negative, thereby ensuring that the feature weights are also non-negative. Notably, the final results for A are determined through the utilization of an l_2 -norm regularizer. Consequently, the feature ranking can be derived, taking into consideration the descending order and the number of selected features.

Algorithm 1 EF²FS

Input: Data matrices $\{X^{(i)}\}_{i=1}^V$; Label matrix Y .

Parameter: Parameters α, β, γ and δ .

Output: Set of selected features.

- 1: Initialize $w^{(i)}, A, B$ and G ;
 - 2: **repeat**
 - 3: Update the matrix $w^{(i)}$ in accordance with the formulation specified in Formula (27);
 - 4: Update the matrix A in accordance with the formulation specified in Formula (25);
 - 5: Update the matrix B in accordance with the formulation specified in Formula (23);
 - 6: Update the matrix G in accordance with the formulation specified in Formula (21);
 - 7: Update the objective function (19);
 - 8: **until** Convergence
 - 9: Derive the ordered feature sequence by performing computations on the weights, denoted as $\|A_{(j)}\|_2$ where $j = 1, 2, 3, \dots, d$.
-

3.3. Optimization

Considering the unsmooth problem caused by $l_{2,1}$ -norm, the term $\|A\|_{2,1}$ needs to be relaxed by introducing another diagonal matrix D , where $d_{ii} = 1/(2\|A_i\|_2)$. The regularizer can be transformed as $2\text{Tr}(A^T D A)$. Meanwhile, the Frobenius norm like $\|E\|_F^2$ can be equivalently expressed as $\text{Tr}(E^T E)$ when the variable E is an arbitrary matrix. Thus, the Formula (17) can be represented with another form:

$$\begin{aligned} \min_{A, G, B, w^{(i)}} & \text{Tr}[(X^f A - G)^T (X^f A - G)] + \alpha \text{Tr}[(Y - GB^T)^T (Y - GB^T)] \\ & + \beta \sum_{i=1}^V \text{Tr}[(x^{(i)} w^{(i)} - G)^T (x^{(i)} w^{(i)} - G)] \\ & + \gamma \sum_{i=1}^V \text{Tr}[(a^{(i)} - w^{(i)})^T (a^{(i)} - w^{(i)})] + 2\delta \text{Tr}(A^T D A). \end{aligned} \quad (18)$$

s.t. $G \geq 0, B \geq 0$

Subsequently, we optimize the objective function of our method through the multiplicative rules. First, the constraints are integrated into the objective function through introducing the Laplacian multiplier $\varphi \in \mathbb{R}^{n \times k}$, $\zeta \in \mathbb{R}^{c \times k}$, $\psi \in \mathbb{R}^{d \times c}$ and $\tau \in \mathbb{R}^{d(i) \times c}$, and the number of multipliers is corresponding to the number of learned variables. The objective function can be written with the Laplacian multiplier as:

$$\begin{aligned} \min_{A, G, B, w^{(i)}} & \text{Tr}[(X^f A - G)^T (X^f A - G)] + \alpha \text{Tr}[(Y - GB^T)^T (Y - GB^T)] \\ & + \beta \sum_{i=1}^V \text{Tr}[(x^{(i)} w^{(i)} - G)^T (x^{(i)} w^{(i)} - G)] \\ & + \gamma \sum_{i=1}^V \text{Tr}[(a^{(i)} - w^{(i)})^T (a^{(i)} - w^{(i)})] + 2\delta \text{Tr}(A^T D A) \\ & - \text{Tr}(\varphi G^T) - \text{Tr}(\zeta B^T) - \text{Tr}(\psi A^T) - \text{Tr}(\tau w^{(i)T}). \end{aligned} \quad (19)$$

Due to the property of non-convex, it is necessary to update the variables individually in a sequential manner, and a detailed breakdown of this process is elucidated in Algorithm 1.

(1) Fix B, A and $w^{(i)}$, update G :

The sub-objective for G can be formulated as follows:

$$\begin{aligned} \Theta_G = & \text{Tr}[(X^f A - G)^T (X^f A - G)] + \alpha \text{Tr}[(Y - GB^T)^T (Y - GB^T)] \\ & + \beta \sum_{i=1}^V \text{Tr}[(x^{(i)} w^{(i)} - G)^T (x^{(i)} w^{(i)} - G)] - \text{Tr}(\varphi G^T). \end{aligned} \quad (20)$$

After obtaining the partial derivatives of Θ with respect to G , and the updating rule of G is written according to the KKT conditions as:

$$G \leftarrow G \circ \frac{\alpha Y B + \beta \sum_{i=1}^V x^{(i)} w^{(i)} + X^f A}{\alpha G B^T B + (\beta V + 1)G}. \quad (21)$$

(2) Fix G, A and $w^{(i)}$, update B :

The sub-objective for B and its updating rule can be formulated as follows:

$$\Theta_B = \alpha \text{Tr}[(Y - GB^T)^T (Y - GB^T)] - \text{Tr}(\zeta B^T). \quad (22)$$

$$B \leftarrow B \circ \frac{Y^T G}{B G^T G}. \quad (23)$$

(3) Fix G, B and $w^{(i)}$, update A :

The sub-objective for A and its updating rule can be formulated as follows:

$$\begin{aligned} \Theta_A = & \text{Tr}[(X^f A - G)^T (X^f A - G)] + \gamma \sum_{i=1}^V \text{Tr}[(a^{(i)} - w^{(i)})^T (a^{(i)} - w^{(i)})] \\ & + 2\delta \text{Tr}(A^T D A) - \text{Tr}(\psi A^T). \end{aligned} \quad (24)$$

Table 2

The detailed information for the datasets in our experiments.

Views	emotions	yeast	SCENE	MIRFlickr	IAPRTC12	3Sources
View-1(d_1)	Rhythmic(8)	GE(79)	CH(64)	DH(100)	DH(100)	BBC(1000)
View-2(d_2)	Timbre(64)	PP(24)	CM(225)	GIST(512)	DHV3H1(300)	Reuters(1000)
View-3(d_3)			CORR(144)	HH(100)	GIST(512)	Guardian(1000)
View-4(d_4)			EDH(73)		HHV3H1(300)	
View-5(d_5)			WT(128)		HH(100)	
Instances(n)	593	2417	4400	4053	4999	169
Features(d)	72	103	634	712	1312	3000
Labels(c)	6	14	33	38	260	6

$$A \leftarrow A \circ \frac{X^f G + \gamma W}{X^f T X^f A + \gamma A + \delta D A}. \quad (25)$$

(4) Fix G , B and A , update $w^{(i)}$:

The sub-objective for $w^{(i)}$ and its updating rule can be formulated as follows:

$$\begin{aligned} \Theta_{w^{(i)}} = & \beta \sum_{i=1}^V \text{Tr} \left[(x^{(i)} w^{(i)} - G)^T (x^{(i)} w^{(i)} - G) \right] \\ & + \gamma \sum_{i=1}^V \text{Tr} \left[(a^{(i)} - w^{(i)})^T (a^{(i)} - w^{(i)}) \right] - \text{Tr}(\tau w^{(i)T}). \end{aligned} \quad (26)$$

$$w^{(i)} \leftarrow w^{(i)} \circ \frac{\beta x^{(i)T} G + \gamma a^{(i)}}{\beta x^{(i)T} x^{(i)} w^{(i)} + \gamma w^{(i)}}. \quad (27)$$

The proof of convergence can be found in [44], and the objective function is non-increasing with a proper auxiliary function $J(b, b')$ for $F(b) = 1/2 \sum_i (y_i - \sum_a G_{ia} b_a^T)^2$, which guarantees $J(b, b') = F(b)$ and $J(b, b') \geq F(b)$.

$$J(b, b') = F(b') + F'(b')(b - b') + \frac{1}{2} F''(b')(b - b')^2, \quad (28)$$

where $F''(b')$ is changed into $K_{ab}(b') = \delta_{ab}(b' G^T G)_a / b'_a$. After the $F(b)$ is expanded with Taylor's theorem, the condition $[K(b') - G^T G](b - b')^2 \geq 0$ holds to prove convergence. Similarly, the objective function is non-increasing regarding other variables.

3.4. Complexity analysis

In the complexity analysis, the main focus lies on four variables G , B , A and $w^{(i)}$, as they collectively determine the overall complexity of the method EF^2FS . To facilitate statistical calculations, we assume a unified dimension value, denoted as d , for different views. Consequently, the complexity associated with each variable can be outlined as:

- (1) Updating G involves five terms of matrix operation, and the complexity for all terms is $(O(nck) + O(ndk))$.
- (2) Updating B involves two terms of matrix operation, and the total computational complexity is $O(nck)$.
- (3) Updating A and $w^{(i)}$ involves five and four terms of matrix operation, yielding a combined computational complexity of $(O(nd^2) + O(d^2k) + O(ndk))$.

Combined with the above, the computational complexity of the whole training process is approximated as $O(nd^2)$ for each iteration, when considering $n \gg d \gg c \gg k$.

4. Experiments**4.1. Experimental setup****4.1.1. Datasets and evaluation metrics**

In our experiments, the datasets adopted in Table 2 are popular in the multi-view multi-label scenario. From Table 2, it can be seen

that the datasets have different number of features as d , that the range of it is from dozens to several thousands. Meanwhile, the number of instances and labels denotes as n and c , respectively. The feature for different views represents various meanings. For the two datasets emotions and yeast, they have two views, which express the field of music and genes. Conversely, in the Scene, MIRFlickr, and IAPRTC12 datasets, which consist of three or five views, each view represents visual properties within the realm of computer vision. It is important to note that the number of views varies across these datasets. For the dataset 3Sources, the views represent different source of the new stories from the period February–April 2009, and the sources are obtained from BBC, Reuters and The Guardian.

To verify the effectiveness for EF^2FS , the evaluation metrics are employed including *Average Precision* (AP), *Coverage*, *Hamming Loss* (HL) and *Ranking Loss* (RL). According to the characteristic of each metric, the principle for the last three metrics inclines to count the number of errors. For example, the ‘errors’ in HL means that the count for absence of predicted labels in the ground-truth set. Thus, the value of these metrics is inverse proportion to the performance. The AP metric is employed to evaluate the precision of the performance, where a higher value signifies better performance. The experimental results are obtained using a five-fold cross-validation, and the mean accuracy and standard deviation for each fold are calculated following a grid search conducted with the range of $\{10^{-3}, 10^{-2}, \dots, 10^3\}$. The methods are performed adhering to these established standards, in order to ensure the validity of the comparative results presented in the paper.

4.1.2. Comparing methods

Six prominent methods are adopted for comparative analysis with our proposed method, which both focus on the feature space. Concurrently, our research aims to substantiate the advantages of our algorithm by assessing its performance across three critical dimensions: feature fusion, paradigm design that prioritizes feature data, and design that emphasizes. In order to conduct a comprehensive evaluation, we meticulously select comparative algorithms that align with these three aspects, enabling an effective comparison. These methods can be categorized into two types. The first type encompasses multi-view multi-label methods such as M2LD [7] and MSFS [14], which prioritize the feature subspace or feature weights. The second type comprises multi-label methods SSFS [24], MoRE [23], MRDM [27] and CLML [33], predominantly focusing on feature selection as the primary objective.

4.2. Experimental results**4.2.1. Feature selection performance**

The performance of the results with four evaluation metrics are listed in Tables 3–4. In these tables, each value represents the average across varying percentages of features. The range of percentages is set from 1 to 20. Through the results above, the method EF^2FS shows its superiority when compared with others, and the clear illustration for the trend of the performance is represented in Fig. 2. To comprehensively evaluate the performance of our method, we utilize the Friedman test to assess the significance level characteristics. Table 5 presents

Table 3Experimental results (mean $\pm$ std) in terms of AP and Coverage, where the 1st/2nd best results are shown in boldface/underline.

Dataset	EF^2FS	M2LD	MSFS	SSFS	MoRE	MRDM	CLML
AP $\uparrow$							
emotions	0.6911 $\pm$ 0.017	0.5906 $\pm$ 0.016	0.604 $\pm$ 0.025	0.6435 $\pm$ 0.049	<u>0.6669 $\pm$ 0.047</u>	0.6059 $\pm$ 0.04	0.6358 $\pm$ 0.018
yeast	0.6639 $\pm$ 0.019	0.6587 $\pm$ 0.015	0.6495 $\pm$ 0.012	0.6555 $\pm$ 0.014	0.6552 $\pm$ 0.016	<u>0.6595 $\pm$ 0.016</u>	0.6496 $\pm$ 0.012
SCENE	0.7935 $\pm$ 0.007	0.7792 $\pm$ 0.014	0.7887 $\pm$ 0.008	0.7774 $\pm$ 0.009	0.7877 $\pm$ 0.007	<u>0.7929 $\pm$ 0.01</u>	0.787 $\pm$ 0.008
MIRFlickr	0.6759 $\pm$ 0.012	0.6436 $\pm$ 0.013	0.6417 $\pm$ 0.01	<u>0.6719 $\pm$ 0.013</u>	0.6322 $\pm$ 0.011	0.6407 $\pm$ 0.008	0.6442 $\pm$ 0.007
IAPRTC12	0.1509 $\pm$ 0.01	0.137 $\pm$ 0.006	0.1385 $\pm$ 0.005	<u>0.1447 $\pm$ 0.006</u>	0.1263 $\pm$ 0.006	0.14 $\pm$ 0.005	0.1352 $\pm$ 0.002
3Sources	0.4608 $\pm$ 0.044	0.4368 $\pm$ 0.038	0.4252 $\pm$ 0.035	<u>0.4385 $\pm$ 0.046</u>	–	0.4196 $\pm$ 0.041	0.4232 $\pm$ 0.046
Coverage $\downarrow$							
emotions	0.5707 $\pm$ 0.013	0.6493 $\pm$ 0.028	0.6779 $\pm$ 0.029	0.609 $\pm$ 0.051	<u>0.5928 $\pm$ 0.046</u>	0.6482 $\pm$ 0.041	0.6193 $\pm$ 0.018
yeast	0.6266 $\pm$ 0.03	0.6425 $\pm$ 0.017	0.6375 $\pm$ 0.021	0.6428 $\pm$ 0.018	<u>0.6357 $\pm$ 0.022</u>	0.6426 $\pm$ 0.023	0.6407 $\pm$ 0.019
SCENE	0.4244 $\pm$ 0.014	0.4416 $\pm$ 0.016	0.4271 $\pm$ 0.012	0.4481 $\pm$ 0.011	0.4295 $\pm$ 0.01	<u>0.4252 $\pm$ 0.017</u>	0.4308 $\pm$ 0.012
MIRFlickr	0.5774 $\pm$ 0.008	0.6069 $\pm$ 0.009	0.6173 $\pm$ 0.013	<u>0.5829 $\pm$ 0.013</u>	0.6397 $\pm$ 0.01	0.6128 $\pm$ 0.017	0.6191 $\pm$ 0.015
IAPRTC12	0.4794 $\pm$ 0.016	0.5131 $\pm$ 0.011	0.5098 $\pm$ 0.009	<u>0.5052 $\pm$ 0.018</u>	0.5877 $\pm$ 0.02	0.5097 $\pm$ 0.017	0.5149 $\pm$ 0.005
3Sources	0.6251 $\pm$ 0.035	<u>0.6397 $\pm$ 0.03</u>	0.6602 $\pm$ 0.037	0.6602 $\pm$ 0.055	–	0.6723 $\pm$ 0.042	0.6727 $\pm$ 0.046

Table 4Experimental results (mean $\pm$ std) in terms of HL and RL, where the 1st/2nd best results are shown in boldface/underline.

Dataset	EF^2FS	M2LD	MSFS	SSFS	MoRE	MRDM	CLML
HL $\downarrow$							
emotions	0.2415 $\pm$ 0.01	0.2948 $\pm$ 0.013	0.3108 $\pm$ 0.025	0.2888 $\pm$ 0.025	<u>0.2815 $\pm$ 0.028</u>	0.3057 $\pm$ 0.014	0.3093 $\pm$ 0.005
yeast	0.2225 $\pm$ 0.005	0.2341 $\pm$ 0.002	0.2299 $\pm$ 0.004	0.232 $\pm$ 0.004	<u>0.2259 $\pm$ 0.006</u>	0.2358 $\pm$ 0.007	0.231 $\pm$ 0.004
SCENE	0.0978 $\pm$ 0.004	0.104 $\pm$ 0.006	0.0999 $\pm$ 0.005	0.1058 $\pm$ 0.005	0.1007 $\pm$ 0.005	<u>0.0979 $\pm$ 0.005</u>	0.101 $\pm$ 0.005
MIRFlickr	0.1744 $\pm$ 0.005	0.2029 $\pm$ 0.007	0.1944 $\pm$ 0.002	<u>0.1773 $\pm$ 0.008</u>	0.1931 $\pm$ 0.003	0.1946 $\pm$ 0.003	0.1917 $\pm$ 0.003
IAPRTC12	0.01549 $\pm$ 0	0.01653 $\pm$ 0	0.01561 $\pm$ 0	0.01574 $\pm$ 0	<u>0.0155 $\pm$ 0</u>	0.0156 $\pm$ 0	0.01554 $\pm$ 0
3Sources	0.2332 $\pm$ 0.026	0.2491 $\pm$ 0.025	<u>0.2357 $\pm$ 0.017</u>	0.2514 $\pm$ 0.021	–	0.2358 $\pm$ 0.017	0.2471 $\pm$ 0.018
RL $\downarrow$							
emotions	0.2732 $\pm$ 0.016	0.3887 $\pm$ 0.033	0.3898 $\pm$ 0.023	0.3295 $\pm$ 0.066	<u>0.3099 $\pm$ 0.06</u>	0.3791 $\pm$ 0.051	0.351 $\pm$ 0.02
yeast	0.2547 $\pm$ 0.015	0.2605 $\pm$ 0.012	0.2614 $\pm$ 0.01	0.2593 $\pm$ 0.01	<u>0.2563 $\pm$ 0.017</u>	0.2574 $\pm$ 0.015	0.2628 $\pm$ 0.01
SCENE	0.0947 $\pm$ 0.007	0.1037 $\pm$ 0.009	0.097 $\pm$ 0.006	0.1056 $\pm$ 0.005	0.0974 $\pm$ 0.005	<u>0.0949 $\pm$ 0.007</u>	0.098 $\pm$ 0.006
MIRFlickr	0.1591 $\pm$ 0.007	0.1672 $\pm$ 0.008	0.1855 $\pm$ 0.006	<u>0.1619 $\pm$ 0.009</u>	0.1925 $\pm$ 0.006	0.1858 $\pm$ 0.007	0.1844 $\pm$ 0.006
IAPRTC12	0.1922 $\pm$ 0.01	0.207 $\pm$ 0.009	0.2093 $\pm$ 0.006	<u>0.2064 $\pm$ 0.012</u>	0.2452 $\pm$ 0.01	0.2069 $\pm$ 0.013	0.2134 $\pm$ 0.003
3Sources	0.4862 $\pm$ 0.042	<u>0.5045 $\pm$ 0.039</u>	0.5253 $\pm$ 0.043	0.5172 $\pm$ 0.06	–	0.536 $\pm$ 0.051	0.5345 $\pm$ 0.055

Table 5The critical value at the significance level $\alpha = 0.05$ and Friedman statistic F_F on four evaluation metrics.

Evaluation metrics	F_F	Critical value ($\alpha = 0.05$)
Average precision	4.368	2.53
Coverage	3.435	
Hamming loss	4.912	
Ranking loss	4.000	

the critical values at the specified significance level and the Friedman statistic for each evaluation metric. The Friedman test operates under the assumption that all methods exhibit equivalent performance. A metric value exceeding the critical value indicates that the null hypothesis is rejected, thus signifying significant differences between methods. As depicted in Table 5, each metric surpasses the critical value of 2.53.

- (1) The advantages of our method are more obvious in the cases. From the observations, it is evident that these methods commonly ranked second, namely SSFS and MoRE, outperform others in approximately 33.4% of the cases. This superior performance is largely due to the effective harnessing of information from the feature space. In this paper, we also procure comprehensive information from both global and local feature spaces.

- (2) According to the frequency of achieving the second ranking, the performance of MRDM is moderately ranked. Contrastingly, CLML demonstrates relative shortcomings when compared to MRDM. This is primarily due to its neglect of non-linear constraints, which leads to its failure to exhibit significant effects among the seven methods discussed in this paper. It can be seen that relying solely on labels makes it challenging to achieve optimal performance. In EF^2FS , we consider label noise alongside a focused investigation of the feature space.

- (3) Upon accommodating the disparities between views, a significant enhancement in performance is discernible. For instance, the multi-view methods exhibit superior performance compared to CLML. Regarding the issue of the objective function in M2LD and MSFS, it is observed that the performance of fusion for the feature matrix as EF^2FS is more readily optimized.

4.2.2. Parameter analysis

The experiment for the sensitivity of four parameters revolves around two dimensions including the number of features and the range of value for parameters. Through observing the evaluation metric for AP in Fig. 3, it can be seen that the method EF^2FS is relatively stable. On the SCENE dataset, the performance consistently exhibits a higher level of stability compared to other datasets, no matter how the variable change. The effect of the range of value for the parameters appears on the MIRFlickr, it can be seen that the precision is changed within

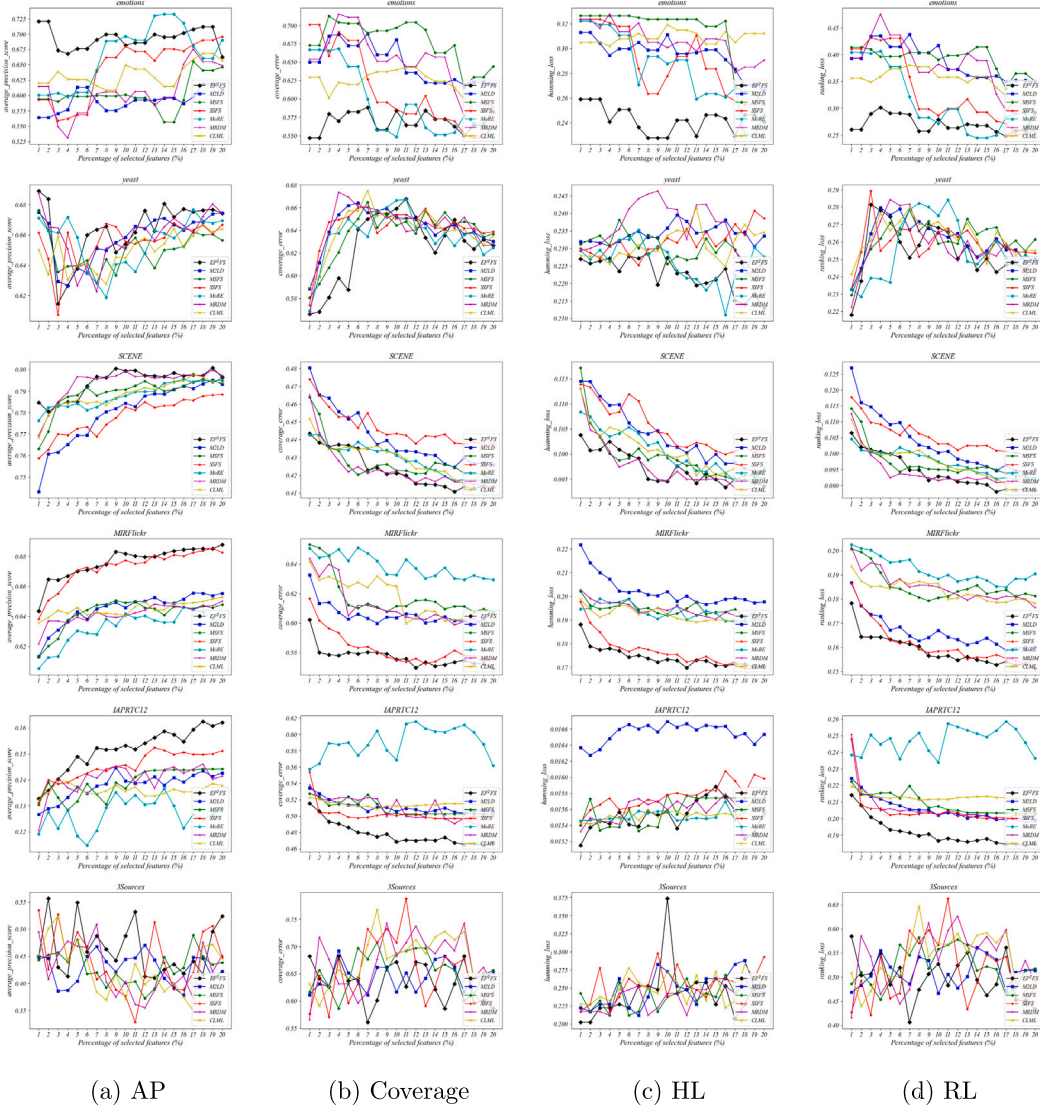


Fig. 2. The performance for the feature selection is characterized on SCENE, MIRFlicker and IAPRTC12 with four evaluation metrics.

a small range, and the maximum amplitude is not beyond 14%. The phenomenon occurs on the IAPRTC12 dataset especially for γ and δ that the performance is improved with the increasing number of features and the magnitude is around 18%, which is also in accordance with the rules of feature selection.

4.2.3. Convergence

The trend of the curve can prove the convergence of the method EF²FS. The Y-axis denotes the converging ratio $(Z^{t-1} - Z^t) / Z^{t-1}$ in Fig. 4. It can be seen that the fast converging at first appears on MIRFlicker and IAPRTC12, which is below the line of 0.2. And then the curve of MIRFlicker is still keeping the speed to achieve the stop criterion. The converging ratio of IAPRTC12 is slightly down corresponding to the bouncing of the curve. Although the curve of SCENE is not decreasing fast, it uses fewer iterations to the end. Overall, the method has been demonstrated to exhibit convergence and possesses a relatively rapid rate of convergence.

4.2.4. Running time comparison

Apart from the complexity of EF²FS, the computational complexities of M2LD, MSFS, SSFS, MoRE, MRDM and CLML is $O(k_2dn)$, $O(ndc + d^2 + n^2)$, $O(nda + n^2a)$, $O(n^2d + n^2\log(n))$, $O(n^3)$ and $O(nd^2 + d^2l + ndl)$ respectively. Here, k_2 represents the dimension of the latent subspace,

a signifies the number of already-selected features, and c denotes the number of label clusters. As illustrated in Table 6, the proposed method ranks within the top three across most datasets. Conversely, MRDM exhibits the slowest running time due to its high complexity, except in cases where the number of instances is minimal, such as in the 3Sources dataset. The complexity of CLML is comparable to that of GRAFS, while MSFS demonstrates higher complexity when there is a substantial disparity between n and d .

4.2.5. Discussion

In order to validate the significance of feature fusion from a global perspective and the effectiveness of feature enhancement as discussed in Section 3, we propose two types of transformations for the original function. It is important to note that the function of feature fusion operates on the loss function of (EF²FS), its functionality cannot be verified by simply removing relevant elements, as done in other ablation studies. The objective of demonstrating the significance of fusion is to investigate whether assigning higher prominence to distinctive features leads to improved performance.

The alternative method derived from EF²FS, namely EF²FS_{ver2}, primarily focuses on investigating the selection of the principal component. In EF²FS, the principal component serves as the global feature weight denoted by A . In order to preserve the distinctive characteristics

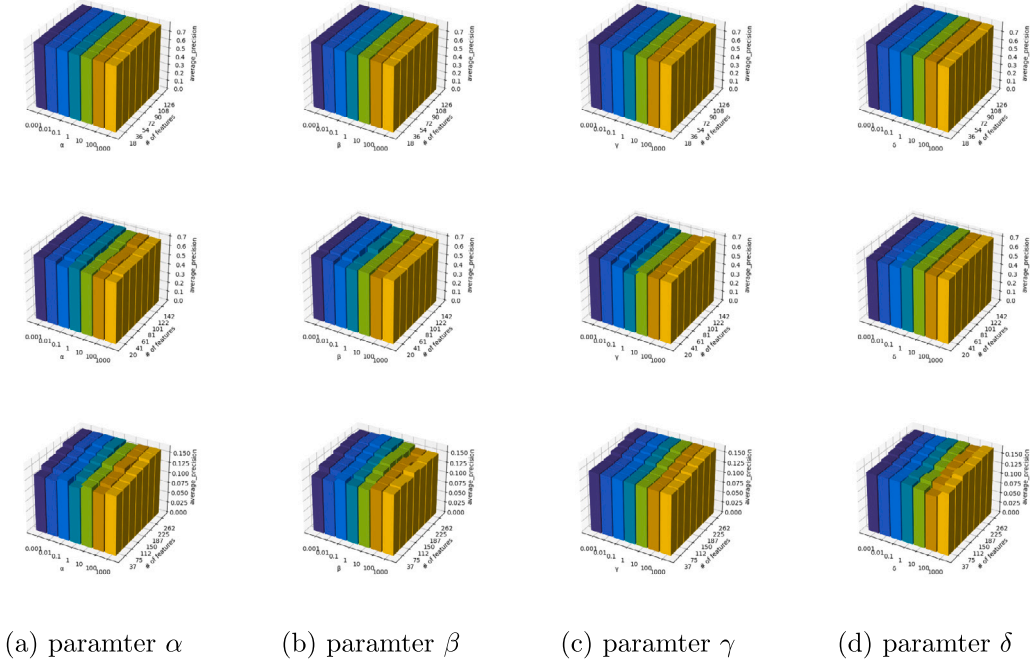


Fig. 3. Parameter sensitivity studies on the SCENE, MIRFlickr and IAPRTC12 datasets.

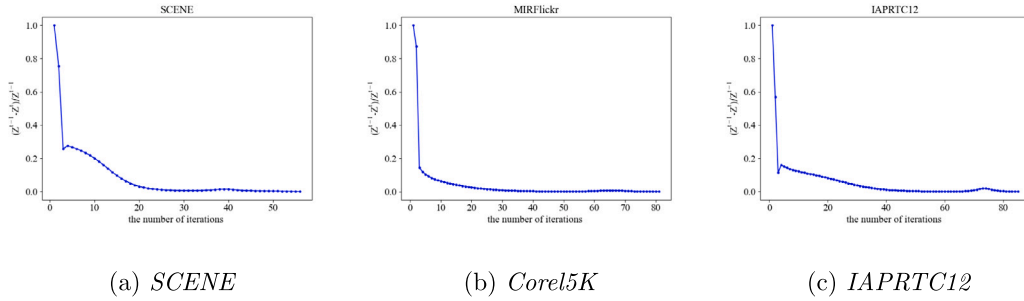


Fig. 4. Convergence curves on SCENE, MIRFlickr and IAPRTC12.

Table 6

Running time (in second) of different methods.

Running time (s)	EF^2FS	M2LD	MSFS	SSFS	MoRE	MRDM	CLML
emotions	0.2536	0.7126	1.8171	0.8818	0.9193	4.6435	1.1851
yeast	2.059	4.903	17.072	1.590	2.918	93.335	6.506
SCENE	23.058	14.337	120.570	9.042	43.108	856.762	91.750
MIRFlickr	17.401	17.956	26.707	10.729	42.756	219.514	98.398
IAPRTC12	54.465	13.653	155.750	19.303	355.308	863.724	613.996
3Sources	11.145	0.745	0.892	2.258	—	5.995	91.691

of each view, the local feature weight is also incorporated into the global feature weight. The question arises as to what would occur if we were to exchange the roles of these two components, such that the principal component is replaced by the local feature weight W , while the global feature weight is utilized to compensate for any deficiencies for locals. Consequently, the method is reformulated, and the objective function EF^2FS_{ver2} is represented as:

$$\begin{aligned} \min_{A, G, B, w(i)} & \sum_{i=1}^V \left\| x^{(i)} w^{(i)} - G \right\|_F^2 + \alpha \left\| Y - G B^T \right\|_F^2 + \beta \left\| X^f A - G \right\|_F^2 \\ & + \gamma \sum_{i=1}^V \left\| a^{(i)} - w^{(i)} \right\|_F^2 + \delta \|W\|_{2,1} . \\ \text{s.t. } & G \geq 0, B \geq 0 \end{aligned} \quad (29)$$

To ensure a fair comparison, all parameters were set to 1, consistent with EF^2FS , and the performance is evaluated through the four metrics. The experimental results are tableted in Table 7, indicating that EF^2FS

outperforms EF^2FS_{ver2} in most of datasets. While EF^2FS does not win on the IAPRTC12 dataset for three metrics, the gap is relatively small, with an approximate 3.88% for AP metric, which is the largest gap observed among the metrics, while the smallest gap is approximately 2%.

Subsequently, another issue arises concerning the usefulness of introducing the distinctive part if the efficacy of designating it as the principal component proves to be inferior. To further investigate the impact of the distinctive part, the parameters β and γ in the objective function are set to 0, rendering the function for preserving distinctiveness ineffective. Consequently, a redefined objective function, denoted as EF^2FS_{ver3} , is formulated as follows:

$$\begin{aligned} \min_{A, G, B, w(i)} & \left\| X^f A - G \right\|_F^2 + \alpha \left\| Y - G B^T \right\|_F^2 + \delta \|A\|_{2,1} . \\ \text{s.t. } & G \geq 0, B \geq 0 \end{aligned} \quad (30)$$

Table 7

The performance of two methods EF^2FS and EF^2FS_{ver2} with different principal part for four evaluation metrics.

AP	emotions	yeast	SCENE	MIRFlickr	IAPRTC12	3Sources
EF^2FS	0.6911	0.6639	0.7935	0.6518	0.1418	0.4608
EF^2FS_{ver2}	0.6504	0.6582	0.7843	0.6452	0.1473	0.4557
Coverage	emotions	yeast	SCENE	MIRFlickr	IAPRTC12	3Sources
EF^2FS	0.5707	0.6266	0.4244	0.6115	0.4966	0.6251
EF^2FS_{ver2}	0.6053	0.6307	0.4326	0.6237	0.4869	0.6320
HL	emotions	yeast	SCENE	MIRFlickr	IAPRTC12	3Sources
EF^2FS	0.2415	0.2225	0.0978	0.1869	0.0157	0.2332
EF^2FS_{ver2}	0.2783	0.2232	0.1020	0.1923	0.0157	0.2334
RL	emotions	yeast	SCENE	MIRFlickr	IAPRTC12	3Sources
EF^2FS	0.2732	0.2547	0.0947	0.1783	0.2030	0.4862
EF^2FS_{ver2}	0.3221	0.2542	0.0993	0.1842	0.1957	0.4893

Table 8

The performance of two methods EF^2FS and EF^2FS_{ver3} with four evaluation metrics on the emotions, yeast and IAPRTC12 dataset.

emotions	AP	Coverage	HL	RL
EF^2FS	0.6911	0.5707	0.2415	0.2732
EF^2FS_{ver3}	0.6883	0.5671	0.2466	0.2743
yeast	AP	Coverage	HL	RL
EF^2FS	0.6639	0.6266	0.2225	0.2547
EF^2FS_{ver3}	0.6582	0.6300	0.2229	0.2529
IAPRTC12	AP	Coverage	HL	RL
EF^2FS	0.1418	0.4966	0.0157	0.2030
EF^2FS_{ver3}	0.1382	0.5117	0.0156	0.2109

Through the performance of the results in Table 8, it is evident that the effect of EF^2FS is obviously better than EF^2FS_{ver3} in most cases, which proves the benefit of incorporating the regularizer in case of ignoring the distinctive part. To summarize, the experimental results effectively showcase the importance of each part and the irreplaceable role they play in the objective function of EF^2FS . These findings reaffirm the critical contributions of the design for fusion and distinctive part.

5. Conclusion

In this paper, the method EF^2FS is proposed from a novel perspective to consider the feature selection problem in MVML scenario. The method not only provides insights into feature fusion to solve the shared space, but also circles around the feature weight to construct the regularizer combined with global view and local view. This enables it to achieve the better effect of balancing the commonality and complementarity. Furthermore, we explore the importance of the global view compared to the local view, as well as the necessity of including the term for the local view. The experimental results prove our superiority for the rationality of the objective function and the performance of feature selection. We will continue to enhance the performance of our methods by reducing the number of parameters required.

CRediT authorship contribution statement

Pingting Hao: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Wanfu Gao:** Writing – review & editing, Supervision, Resources. **Liang Hu:** Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work is funded by: by Science Foundation of Jilin Province of China under Grant No. 20230508179RC, and China Postdoctoral Science Foundation funded project under Grant No. 2023M731281.

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